Arunabha Majumder

+45 71 62 24 00 | Aalborg, Denmark
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder

The Exploration Company, Munich

Mechanism Design Engineer

Dear Exploration Company team,

I'd like to apply for the Mechanism Design Engineer role. Designing mechanisms, working out the kinematics and torques by hand and then proving them on real hardware is the part of engineering I enjoy most, and doing that for Nyx would be a dream.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton. The core is a variable stiffness mechanism with a worm gear that pretensions the spring, working in parallel with a 6 Nm motor, and together they give the torque needed to lift the arm. I optimised the mechanism with a genetic algorithm (diameter, spring stiffness and pretension force), made the CAD and drawings in SolidWorks, 3D printed the first parts and had the final ones manufactured. I integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance. The force estimation came from the variable stiffness actuators and the robot's Jacobian, so kinematic and force calculations are something I do every week. I ran the arm impedance study with this robot on 6 participants, with Qualisys cameras tracking the arm and an Arduino-based trigger I implemented to keep the robot and the recording in sync.

Before that, in my Master's at CSIR-CMERI, I fabricated pneumatic artificial muscles for a soft exosuit and designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and LabVIEW. I have written and presented papers on both projects. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), so clear sketches and figures for design reviews come naturally to me.

I'm ready to move to Munich and I'm willing to learn German. I would love to bring new and innovative ideas to your mechanisms and grow into the space side of the work with your team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder